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Ranjit Das

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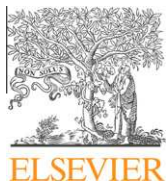
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Magnitude conversion to unified moment magnitude using orthogonal regression relation

Ranjit Das*, H.R. Wason, M.L. Sharma

Department of Earthquake Engineering, Indian Institute of Technology Roorkee, Roorkee 247 667, India

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ABSTRACT

Homogenization of earthquake catalog being a pre-requisite for seismic hazard assessment requires region based magnitude conversion relationships. Linear Standard Regression (SR) relations fail when both the magnitudes have measurement errors. To accomplish homogenization, techniques like Orthogonal Standard Regression (OSR) are thus used. In this paper a technique is proposed for using such OSR for preparation of homogenized earthquake catalog in moment magnitude M_w .

For derivation of orthogonal regression relation between m_b and M_w , a data set consisting of 171 events with observed body wave magnitudes ($m_{b,obs}$) and moment magnitude ($M_{w,obs}$) values has been taken from ISC and GCMT databases for Northeast India and adjoining region for the period 1978–2006. Firstly, an OSR relation given below has been developed using $m_{b,obs}$ and $M_{w,obs}$ values corresponding to 150 events from this data set.

$$M_w = 1.3(\pm 0.004)m_{b,proxy} - 1.4(\pm 0.130),$$

where $m_{b,proxy}$ are body wave magnitude values of the points on the OSR line given by the orthogonality criterion, for observed ($m_{b,obs}$, $M_{w,obs}$) points. A linear relation is then developed between these 150 $m_{b,obs}$ values and corresponding $m_{b,proxy}$ values given by the OSR line using orthogonality criterion. The relation obtained is

$$m_{b,proxy} = 0.878(\pm 0.03)m_{b,obs} + 0.653(\pm 0.15).$$

The accuracy of the above procedure has been checked with the rest of the data i.e., 21 events values. The improvement in the correlation coefficient value between $m_{b,obs}$ and M_w estimated using the proposed procedure compared to the correlation coefficient value between $m_{b,obs}$ and $M_{w,obs}$ shows the advantage of OSR relationship for homogenization.

The OSR procedure developed in this study can be used to homogenize any catalog containing various magnitudes (e.g., M_L , m_b , M_S) with measurement errors, by their conversion to unified moment magnitude M_w . The proposed procedure also remains valid in case the magnitudes have measurement errors of different orders, i.e. the error variance ratio is different from unity.

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1. Introduction

Magnitude of an earthquake is the most commonly used descriptor for earthquake size. Because of the inherent complex nature of the earthquake phenomena and variations in instrumentation characteristics for observation of seismic waves at different epicentral distances, most magnitude determinations are subject to measurement errors. Local magnitude (M_L), surface wave magnitude (M_S), body wave magnitude (m_b) and duration magnitude (M_D) are the magnitudes most commonly reported in seismic catalogs. Most widely reported magnitude scale is the local magnitude or Richter

scale magnitude defined by [Richter \(1935\)](#). The surface wave magnitude and the body wave magnitude were proposed later. Based on empirical and theoretical evidences, the differences in the source mechanism, size and duration produce different relative values for the amplitudes of surface and body waves ([Prozorov and Hudson, 1974](#)). In general, the ratio of the body to surface wave amplitude may fluctuate significantly from one earthquake to another. It is necessary to use statistical methods treating earthquakes with the same general source condition in large samples for the reliable detection of a systematic trend.

Uncertainties associated with different magnitude scales play an important role during conversion of the magnitudes into a unified magnitude scale. Very few studies have been devoted towards quantifying these uncertainties in the heterogeneous

* Corresponding author.

E-mail address: ranjit244614@gmail.com (R. Das).

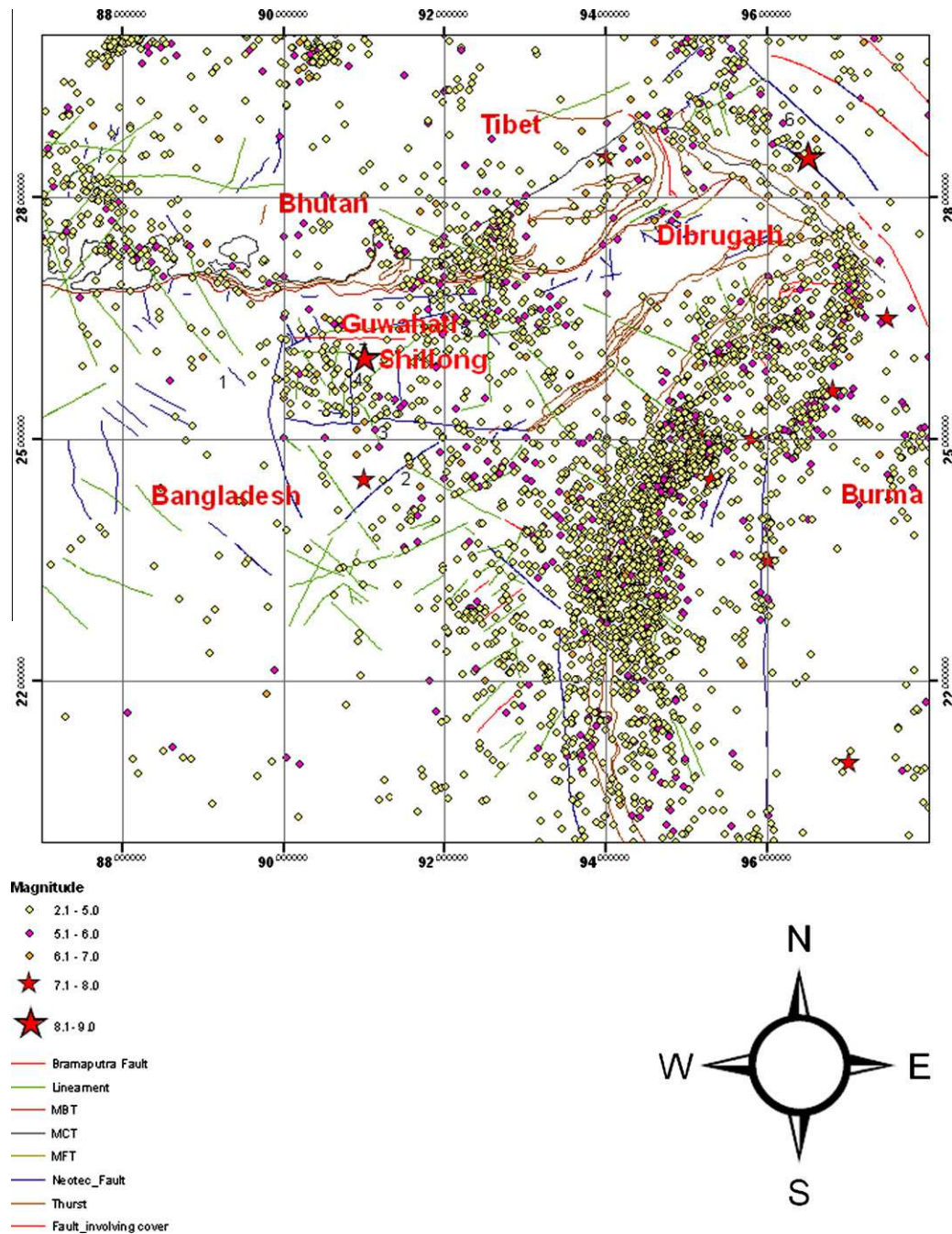


Fig. 1. A seismotectonic map of Northeast India and adjoining region on GIS platform depicting seismicity for $M_w \geq 3.0$.

magnitude scales and their scaling relations. Also the conversion used to compensate for parameter incompatibilities is usually an aleatory variability since the two parameters of magnitude scales, viz., two scales are generally not perfectly correlated (Bommer et al., 2005). The empirically obtained curves for the relationship between two magnitude scales represent earthquakes with average source characteristics (Utsu, 2002). The data for developing such relationships between two magnitude scales are generally very less and give a selected sample choice to obtain a relationship between two magnitude scales for the region under study.

For studies related to earthquake catalogs, it is pertinent to know how different magnitude scales compare with each other

and the order of measurement errors associated with them. The errors in the determination of different magnitudes, in general, differ from each other (Kagan, 2003). Since the error variances in different magnitude types are in general not available, it is widely in use to obtain regression conversions following standard linear least squares approach (Gasperini and Ferrari, 2000; Gasperini, 2002; Bindi et al., 2005). This approach is based on the assumption that the independent variable must be known to a much greater accuracy than the dependent variable. Further, the errors in the dependent variable are taken to be independent and normally distributed. Many empirical relationships have been developed between various magnitude scales for mapping one magnitude type

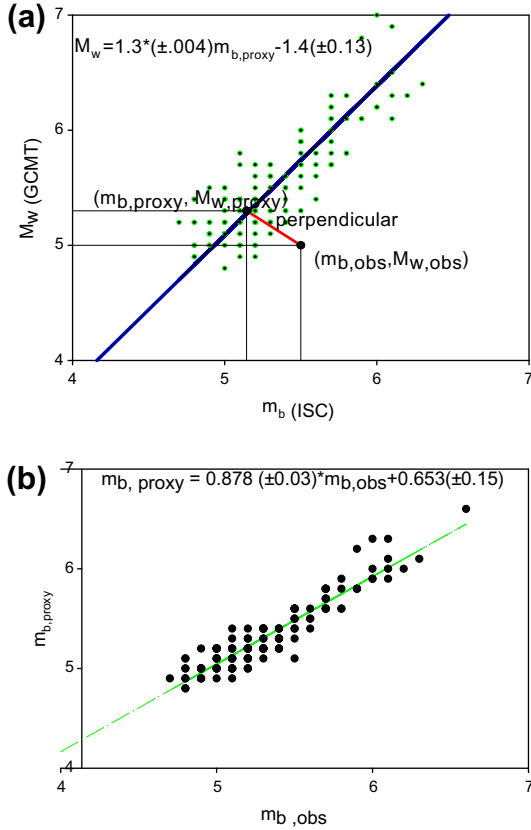


Fig. 2. (a) Orthogonal standard regression relation for $m_{b,ISC}$ and $M_{w,GCMT}$ showing the observed point $(m_{b,obs}, M_{w,obs})$ and the corresponding point $(m_{b,proxy}, M_{w,proxy})$ on the OSR line. (b) Linear Standard Regression between $m_{b,obs}$ and $m_{b,proxy}$ values.

into the other (Gutenberg and Richter, 1956; Bath, 1968; Marshall, 1970; Gibowicz, 1972; Das and Wason, 2010; Nguyen et al., 2011) for different regions in the world.

The use of least-squares linear regression procedure may lead to incorrect results due to both the magnitudes having measurement errors. In such a situation, it is appropriate to use orthogonal regression procedure which takes into account the errors on both the magnitudes (Stromeyer et al., 2004; Joshi and Sharma, 2006; Thingbaijam et al., 2008; Ristau, 2009; Das et al., 2011). However, this regression procedure requires the knowledge of the error variance ratio for the two magnitudes, which is usually not known. An alternative to this problem is to take the error variance ratio equal to unity, assuming that error variance of different magnitudes are approximately equal (Ambraseys, 1990; Gusev, 1991; Panza et al., 1993; Cavallini and Rebez, 1996; Kaverina et al., 1996; Gutdeutsch et al., 2002; Stromeyer et al., 2004).

In this paper, we discuss the use of such orthogonal regression for conversion of body wave magnitude to moment magnitude. In this regard, an Orthogonal Standard Regression (OSR) relation has been developed based on a data set of 150 events from ISC and GCMT databases for Northeast India and adjoining region. Validity of the proposed procedure has been tested on a separate data set of 21 events. The proper use of such orthogonally regressed two magnitude scales lead to more realistic seismic hazard assessment.

2. Orthogonal standard regression

Orthogonal standard regression is used to account for the effects of measurement error in both the variables. The details of orthogo-

nal standard regression procedure are given in the literature (Madansky, 1959; Kendall and Stuart, 1979; Fuller, 1987; Carroll and Ruppert, 1996; Castellaro et al., 2006; Das et al., 2011) and a brief description is given below.

Let M_x and M_y be the true values and m_x and m_y be the observed values with δ and ε as measurement errors of the independent and dependent variable, magnitudes, respectively, then we may write

$$m_x = M_x + \delta, \quad (1)$$

$$m_y = M_y + \varepsilon, \quad (2)$$

and the regression-like model as

$$m_y = \alpha + \beta M_x + \varepsilon, \quad (3)$$

with $M_y = \alpha + \beta M_x$, where α and β are the slope and intercept of the linear relation between the true values.

In the above relations, M_x , ε and δ are taken to be distributed normally and independently.

The covariances of observed values $\sigma_{m_y}^2$, $\sigma_{m_x m_y}$ and $\sigma_{m_x}^2$ are given by

$$\sigma_{m_y}^2 = \beta^2 \sigma_{M_x}^2 + \sigma_{\varepsilon}^2, \quad (4)$$

$$\sigma_{m_x m_y} = \beta \sigma_{M_x}^2, \quad (5)$$

$$\sigma_{m_x}^2 = \sigma_{M_x}^2 + \sigma_{\delta}^2, \quad (6)$$

where the error variance ratio

$$\eta = \frac{\sigma_{\varepsilon}^2}{\sigma_{\delta}^2} \quad (7)$$

If, $s_{m_y}^2$, $s_{m_x}^2$ and $s_{m_x m_y}$ are the sample covariances of m_y , m_x and between m_y and m_x , then

$$s_{m_y}^2 = \hat{\beta}^2 \hat{\sigma}_{M_x}^2 + \eta \hat{\sigma}_{\delta}^2, \quad (8)$$

$$s_{m_x m_y} = \hat{\beta} \hat{\sigma}_{M_x}^2, \quad (9)$$

$$s_{m_x}^2 = \hat{\sigma}_{M_x}^2 + \hat{\sigma}_{\delta}^2. \quad (10)$$

From the above simultaneous Eqs. (8)–(10), the estimators $\hat{\beta}^2$, $\hat{\sigma}_{M_x}^2$ and $\hat{\sigma}_{\delta}^2$ can be easily determined. For example, on elimination of $\hat{\sigma}_{M_x}^2$ and $\hat{\sigma}_{\delta}^2$, we get the quadratic equation

$$\hat{\beta}^2 s_{m_x m_y} - \hat{\beta} (s_{m_y}^2 - \eta s_{m_x}^2) - \eta s_{m_x m_y} = 0,$$

which yields

$$\hat{\beta} = \frac{s_{m_y}^2 - \eta s_{m_x}^2 + \sqrt{(s_{m_y}^2 - \eta s_{m_x}^2)^2 + 4\eta s_{m_x m_y}^2}}{2s_{m_x m_y}} \quad (11)$$

The $\hat{\sigma}_{M_x}^2$ and $\hat{\sigma}_{\delta}^2$ are similarly derived as follows:

$$\hat{\sigma}_{M_x}^2 = \frac{\sqrt{(s_{m_y}^2 - \eta s_{m_x}^2)^2 + 4\eta s_{m_x m_y}^2} - (s_{m_y}^2 - \eta s_{m_x}^2)}{2\eta}, \quad (12)$$

$$\hat{\sigma}_{\delta}^2 = \frac{(s_{m_y}^2 + \eta s_{m_x}^2) - \sqrt{(s_{m_y}^2 - \eta s_{m_x}^2)^2 + 4\eta s_{m_x m_y}^2}}{2\eta}. \quad (13)$$

The estimator for α can be obtained from the relation

$$\hat{\alpha} = \bar{m}_y - \hat{\beta} \bar{m}_x \quad (14)$$

where $\bar{m}_x$ and $\bar{m}_y$ are the average observed values.

Using equations given in Fuller (1987), the estimated variances of regression parameters $\hat{\beta}$ and $\hat{\alpha}$, can be represented as follows:

Table 1

Data for 150 events with observed $m_{b,ISC}$ and $M_{w,GCMF}$ along with the estimated $m_{b,proxy}$ and M_w values from the OSR line for Northeast India and adjoining region. The header abbreviations stand for MO – Month, DD – Day, HH – hour, MM – minute, SS – second, Lat – latitude, Long – longitude.

Event no.	Year	MO	DD	HH	MM	SS	Lat (°N)	Long (°E)	$m_{b,obs}$	$M_{w,obs}$	Point on OSR line corresponding to ($m_{b,obs}$, $M_{w,obs}$)	
											$m_{b,proxy}$	M_w
1	1978	2	23	23	18	34	23.0836	94.7018	5.0	5.1	5.0	5.1
2	1979	1	1	18	51	10.9	20.8941	93.6874	5.3	5.3	5.2	5.4
3	1979	5	29	0	39	52.1	24.4963	94.7404	5.2	5.3	5.2	5.3
4	1979	6	19	16	29	8.4	26.7422	87.4817	5.2	5.0	5.0	5.1
5	1979	7	13	23	20	9.9	24.881	95.2226	4.9	5.0	4.9	5.0
6	1979	11	25	2	40	48	25.2068	96.3227	5.0	5.2	5.1	5.2
7	1979	12	21	6	31	52	27.0961	97.0364	5.5	5.3	5.3	5.5
8	1980	2	22	3	2	44.8	30.5515	88.6459	5.7	6.3	5.8	6.2
9	1980	11	19	19	0	44.6	27.4015	88.7972	6.0	6.2	5.9	6.3
10	1980	11	20	18	14	11.4	22.7363	93.9228	5.2	5.2	5.1	5.3
11	1981	4	25	11	32	23	24.8951	95.3415	5.7	5.7	5.6	5.8
12	1981	5	1	4	8	10	22.9415	94.5595	4.8	4.9	4.8	4.9
13	1981	6	30	21	55	49.8	22.5004	95.189	5.1	4.9	4.9	5.0
14	1981	8	14	6	9	34.4	25.1462	97.9587	5.2	5.2	5.1	5.3
15	1981	8	16	18	55	42.2	25.5244	96.6313	5.0	5.1	5.0	5.1
16	1981	9	12	5	25	19.9	21.0937	99.3572	4.9	5.1	5.0	5.0
17	1982	1	22	4	29	55.9	30.8911	89.8667	5.3	5.5	5.3	5.5
18	1982	6	15	23	24	28.8	31.8514	99.9221	5.5	5.7	5.5	5.7
19	1983	4	17	23	16	33.8	22.0298	94.3563	5.1	5.0	5.0	5.1
20	1983	8	23	12	12	17.5	24.5459	95.1203	5.2	5.2	5.1	5.3
21	1983	8	30	10	39	27.2	25.0393	94.6695	5.7	5.7	5.6	5.8
22	1983	10	21	8	44	47.3	22.0046	94.38	5.3	5.2	5.2	5.3
23	1983	11	16	0	54	11.4	26.1567	96.1167	5.0	5.4	5.2	5.3
24	1984	3	5	21	26	42.6	24.516	94.6204	5.2	5.4	5.2	5.4
25	1984	4	23	22	29	57.3	22.0546	99.1749	5.7	5.9	5.7	5.9
26	1984	5	6	15	19	11.3	24.2152	93.5256	5.7	6.0	5.7	6.0
27	1984	11	28	10	29	20.5	26.6489	97.0842	5.7	5.7	5.6	5.8
28	1984	12	30	23	33	35	24.6573	92.8468	5.5	6.0	5.6	5.9
29	1985	4	24	6	47	45.2	26.1811	96.0759	5.3	5.2	5.2	5.3
30	1985	8	1	12	13	45.4	29.1522	95.1634	5.4	5.6	5.4	5.6
31	1985	9	5	18	30	21.6	25.3958	97.7055	5.0	5.5	5.2	5.3
32	1986	2	8	0	28	53.5	23.8665	93.0026	5.2	5.4	5.2	5.4
33	1986	2	19	17	34	23.2	25.1044	91.1302	5.2	5.3	5.2	5.3
34	1986	4	26	0	25	58.4	22.8469	94.5113	4.9	5.1	5.0	5.0
35	1986	7	26	20	24	49.6	23.7149	94.1938	5.2	5.4	5.2	5.4
36	1986	11	1	5	2	40.3	26.8483	96.3965	5.4	5.2	5.2	5.4
37	1987	5	18	1	53	51.3	25.2287	94.2076	5.7	6.2	5.8	6.1
38	1987	8	24	9	24	40	23.0498	94.4147	5.1	5.3	5.1	5.3
39	1987	9	25	23	16	29.2	29.8407	90.3668	5.1	5.0	5.0	5.1
40	1988	1	25	1	12	22.2	30.1572	94.8741	5.4	5.2	5.2	5.4
41	1988	2	6	14	50	45.4	24.6677	91.5619	5.8	5.8	5.6	5.9
42	1988	7	3	8	19	18.6	22.0654	94.2643	5.2	5.0	5.0	5.1
43	1988	8	6	0	36	25.5	25.1297	95.1493	6.6	7.2	6.6	7.2
44	1988	8	13	19	59	51	25.2877	95.1289	5.0	5.0	5.0	5.0
45	1988	8	21	13	16	30.2	25.2713	95.1021	4.9	5.2	5.0	5.1
46	1988	9	3	12	52	46.1	29.947	97.3271	5.0	4.8	4.9	4.9
47	1988	10	23	11	43	9.4	20.3025	94.4125	5.1	5.1	5.0	5.1
48	1988	11	6	13	3	19.9	22.7995	99.5944	6.0	7.0	6.3	6.8
49	1988	11	7	2	39	56.3	23.4267	99.4922	4.8	5.2	5.0	5.1
50	1988	11	15	10	28	14.6	23.1452	99.6472	5.0	5.3	5.1	5.2
51	1988	11	27	4	17	56.4	22.7297	99.8582	5.0	5.5	5.2	5.3
52	1988	11	30	8	13	30	22.7614	99.8442	5.5	6.0	5.6	5.9
53	1989	2	3	17	50	0.2	30.187	89.9442	5.4	5.4	5.3	5.5
54	1989	2	12	7	55	45.9	26.2007	96.9009	5.1	5.3	5.1	5.3
55	1989	3	1	3	25	4.8	21.7421	97.975	5.1	5.7	5.3	5.5
56	1989	4	3	19	39	31.5	25.1516	94.6641	5.3	5.6	5.4	5.6
57	1989	4	9	2	31	36.3	29.1128	90.0223	5.1	5.1	5.0	5.1
58	1989	4	13	7	25	33.4	24.4041	92.4312	5.0	5.4	5.2	5.3
59	1989	4	15	20	34	10.1	29.9729	99.2295	6.0	6.4	6.0	6.4
60	1989	4	25	2	13	20.9	30.0085	99.4256	6.1	6.1	5.9	6.3
61	1989	5	3	5	53	2.8	30.0653	99.4947	6.0	6.2	5.9	6.3
62	1989	5	3	15	41	29.7	30.0398	99.528	5.8	5.8	5.6	5.9
63	1989	5	7	0	38	18.6	23.541	99.5371	5.4	5.6	5.4	5.6
64	1989	6	12	0	4	9.8	21.8338	89.7751	5.7	5.8	5.6	5.9
65	1989	7	15	0	9	14.9	22.7882	94.538	5.4	5.5	5.3	5.5
66	1989	7	21	3	9	15.1	30.0153	99.4626	5.5	5.6	5.4	5.7
67	1989	9	24	10	55	20.2	20.6907	94.9483	5.4	5.3	5.3	5.4
68	1989	9	28	21	52	17.4	20.3643	98.8173	5.5	5.8	5.5	5.8
69	1989	9	30	18	19	23.4	20.2342	98.8598	5.3	5.7	5.4	5.6
70	1989	12	2	19	44	26.8	21.2079	93.8188	5.2	5.3	5.2	5.3
71	1989	12	8	0	4	26.7	21.1857	93.7961	5.6	5.4	5.4	5.6
72	1990	1	9	18	51	29.2	24.7439	95.2586	6.1	6.3	6.0	6.4

(continued on next page)

Table 1 (continued)

Event no.	Year	MO	DD	HH	MM	SS	Lat (°N)	Long (°E)	$m_{b,obs}$	$M_{w,obs}$	Point on OSR line corresponding to ($m_{b,obs}$, $M_{w,obs}$)	
											$m_{b,proxy}$	M_w
73	1990	3	8	18	57	0.3	25.4494	96.6386	4.8	5.2	5.0	5.1
74	1991	1	5	14	57	11.2	23.5455	95.9627	6.1	6.9	6.3	6.8
75	1991	5	11	2	15	22.2	24.258	93.6762	5.0	5.4	5.2	5.3
76	1991	12	4	3	27	23.5	23.9714	93.9102	4.9	5.5	5.2	5.3
77	1991	12	20	2	6	5.2	24.6903	93.1177	5.3	5.3	5.2	5.4
78	1992	3	25	22	32	34.2	24.8186	95.2505	5.2	5.1	5.1	5.2
79	1992	3	27	0	5	18.4	20.8693	94.5881	5.3	5.6	5.4	5.6
80	1992	4	15	1	32	11.3	24.268	94.9275	5.5	5.7	5.5	5.7
81	1992	4	23	14	18	35.4	22.4324	98.9257	5.7	6.1	5.8	6.1
82	1992	4	23	15	32	49.3	22.4254	98.8766	5.7	6.1	5.8	6.1
83	1992	6	15	2	48	56.1	24.0044	95.9676	5.8	6.3	5.9	6.2
84	1992	7	8	10	9	47.8	21.0602	93.6803	5.4	5.2	5.2	5.4
85	1992	7	9	21	34	2.1	21.0464	90.0244	5.2	5.3	5.2	5.3
86	1992	7	30	8	24	49.2	29.5658	90.1799	5.8	6.1	5.8	6.1
87	1992	11	22	11	42	45.4	20.33	94.3193	5.3	5.2	5.2	5.3
88	1993	1	18	12	42	4.5	30.8436	90.378	5.7	5.9	5.7	5.9
89	1993	3	20	14	51	59.7	29.027	87.3284	5.7	6.2	5.8	6.1
90	1993	3	20	21	26	39.7	29.0103	87.3403	4.9	5.1	5.0	5.0
91	1993	4	1	16	30	9.8	23.2137	94.4557	5.3	5.1	5.1	5.2
92	1993	7	17	9	46	34.1	27.9898	99.615	5.2	5.4	5.2	5.4
93	1994	1	11	0	51	56.4	25.2131	97.2121	5.9	6.1	5.8	6.2
94	1994	4	6	7	3	24.3	26.1622	96.8417	5.6	5.8	5.6	5.8
95	1994	5	29	14	11	51.4	20.5378	94.1826	6.1	6.5	6.1	6.5
96	1994	8	3	14	59	58.1	21.5029	93.9791	5.6	5.7	5.5	5.8
97	1994	8	8	21	8	31.1	24.7115	95.2123	5.9	6.1	5.8	6.2
98	1994	11	21	8	16	34.1	25.5352	96.6675	5.5	5.9	5.6	5.8
99	1995	2	17	2	44	24.7	27.6369	92.3959	5.1	5.4	5.2	5.3
100	1995	5	6	1	59	7	24.9605	95.2949	6.3	6.4	6.1	6.5
101	1995	5	9	9	54	18.8	25.2425	95.0974	5.1	5.2	5.1	5.2
102	1995	7	9	20	31	32.1	21.9867	99.1703	5.5	5.9	5.6	5.8
103	1995	7	11	21	46	40	21.9756	99.1989	5.9	6.8	6.2	6.6
104	1996	6	9	23	25	15.7	28.3772	92.2606	5.1	5.2	5.1	5.2
105	1996	7	3	6	44	41.5	30.1058	88.191	5.6	5.6	5.5	5.7
106	1996	7	3	10	10	37.7	30.1876	88.2438	4.9	5.0	4.9	5.0
107	1996	7	26	13	9	2.8	25.1073	96.2052	5.2	5.3	5.2	5.3
108	1996	7	27	8	45	17.8	21.3023	94.7757	5.0	5.2	5.1	5.2
109	1996	7	31	8	0	30.7	30.2358	88.2089	5.0	5.4	5.2	5.3
110	1996	11	19	0	12	16	24.567	92.6825	5.4	5.3	5.3	5.4
111	1996	12	21	8	39	40.1	30.6173	99.3853	5.1	5.4	5.2	5.3
112	1997	4	14	17	53	34.7	22.5907	94.4773	4.9	5.2	5.0	5.1
113	1997	5	8	2	53	14.5	24.8899	92.2768	5.5	5.9	5.6	5.8
114	1997	7	11	14	54	50.3	21.7444	94.8557	5.2	5.3	5.2	5.3
115	1997	7	31	15	59	37	23.9099	93.2207	5.3	5.2	5.2	5.3
116	1997	8	9	4	48	2	30.3272	97.0033	5.1	5.2	5.1	5.2
117	1997	11	21	11	23	3.5	22.2195	92.6839	5.9	6.1	5.8	6.2
118	1997	12	30	13	43	19.1	25.4037	96.5903	5.2	5.7	5.4	5.6
119	1998	5	2	8	36	48.8	24.936	95.2615	5.2	5.5	5.3	5.4
120	1998	7	20	1	6	1.2	30.175	88.2454	5.3	5.7	5.4	5.6
121	1998	7	21	14	40	47.9	30.3022	88.2085	4.8	5.0	4.9	4.9
122	1998	8	25	7	41	43.1	30.2724	88.1617	5.1	5.8	5.4	5.6
123	1998	8	28	22	1	55.1	30.2876	88.2246	4.8	5.0	4.9	4.9
124	1998	9	26	18	27	1.3	27.7646	92.8068	5.5	5.0	5.1	5.3
125	1998	9	30	2	29	55.5	30.014	88.1178	4.9	5.1	5.0	5.0
126	1998	10	5	10	24	47.9	30.2623	88.2565	4.7	5.2	4.9	5.0
127	1998	10	16	0	5	31.3	23.7131	94.5969	5.1	5.3	5.1	5.3
128	1999	2	22	11	37	44.9	23.264	93.642	5.1	5.0	5.0	5.1
129	1999	4	5	22	32	52.6	24.935	93.716	5.3	5.5	5.3	5.5
130	1999	10	5	17	4	45	26.293	91.982	5.2	5.2	5.1	5.3
131	2000	1	26	21	37	57.5	30.893	95.487	4.9	5.1	5.0	5.0
132	2000	6	7	21	46	55.9	26.798	97.187	6.2	6.3	6.0	6.4
133	2000	6	8	12	21	10.4	26.632	97.144	5.0	5.1	5.0	5.1
134	2000	7	2	4	27	56	24.515	94.705	5.0	5.2	5.1	5.2
135	2000	10	6	12	5	40.8	24.379	97.797	5.1	5.4	5.2	5.3
136	2000	10	11	9	42	9.5	23.848	94.803	5.4	5.5	5.3	5.5
137	2000	11	13	8	56	49	21.695	92.856	4.8	5.4	5.1	5.2
138	2001	3	3	22	55	56.8	23.901	93.678	5.3	5.2	5.2	5.3
139	2001	4	10	22	8	13.5	24.589	94.932	4.9	5.2	5.0	5.1
140	2001	4	12	10	46	59.8	24.746	99.053	5.3	5.6	5.4	5.6
141	2001	8	12	1	57	58.4	24.452	94.935	4.9	5.0	4.9	5.0
142	2002	8	8	11	42	4.6	30.851	99.889	5.4	5.3	5.3	5.4
143	2002	10	16	1	31	15.2	21.1822	93.4992	5.0	5.1	5.0	5.1
144	2003	3	25	18	51	26.8	27.2558	89.3791	4.8	5.4	5.1	5.2
145	2003	7	26	23	18	17	22.8929	92.3317	5.4	5.6	5.4	5.6
146	2003	7	27	12	7	29	22.8371	92.3359	5.0	5.4	5.2	5.3

Table 1 (continued)

Event no.	Year	MO	DD	HH	MM	SS	Lat (°N)	Long (°E)	$m_{b,obs}$	$M_{w,obs}$	Point on OSR line corresponding to ($m_{b,obs}$, $M_{w,obs}$)	
											$m_{b,proxy}$	M_w
147	2003	8	18	9	3	2.9	29.5475	95.5627	5.5	5.5	5.4	5.6
148	2004	3	7	13	29	44.6	31.6501	91.2206	5.2	5.6	5.3	5.5
149	2004	5	24	22	0	56.1	27.0498	97.1392	4.8	4.9	4.8	4.9
150	2004	9	27	17	5	36.3	29.814	95.5477	5.2	4.9	5.0	5.1

$$\hat{\sigma}_{\beta}^2 = \frac{\hat{\sigma}_{M_x}^2 (n-1)(\eta + \hat{\beta}^2)\hat{\sigma}_{\delta}^2 + (\hat{\sigma}_{\delta}^2)^2 (n-1)(\eta + \hat{\beta}^2) - (n-2)(-\hat{\beta}\hat{\sigma}_{\delta}^2)^2}{(n-2)(n-1)(\hat{\sigma}_{M_x}^2)^2}, \quad (15)$$

and

$$\hat{\sigma}_{\alpha}^2 = \frac{(n-1)(\eta + \hat{\beta}^2)\hat{\sigma}_{\delta}^2}{n(n-2)} + \bar{m}_x^2 \hat{\sigma}_{\beta}^2, \quad (16)$$

where n is the sample size.

3. Study area and data set

The Northeast India and adjoining region bounded by latitude 20–32°N and longitude 87–100°E (see Fig. 1) comprises of a highly active seismic region. The Shillong massif was the seat of the great earthquake of 12th June, 1897. On 15th August, 1950, another great earthquake occurred further Northeast close to the India–China border. The seismicity in this region is related to the collision of the Indian Plate with Tibet towards the north and the Burmese landmass towards the east (e.g., Seeber and Armbruster, 1981; Khattri and Tyagi, 1983; Molnar, 1987; Molnar and Pandey, 1989; Paul and Sharma, 2011). In addition to the two great earthquakes, this region has experienced several large earthquakes ($M \geq 7.0$) that caused loss of human lives and destruction to buildings.

Events data for 171 events from ISC and GCMT databases for Northeast India and adjoining region with body wave magnitude

and moment magnitude for the time period 1978–2006 has been considered. The seismicity of this region for the time period 1897–2006 has been plotted on the tectonic map of the region in Fig. 1.

4. Derivation of regression relationships

An OSR relation between body wave magnitude and moment magnitude has been developed for Northeast India and adjoining region. For the purpose of validation of the developed relationship, the data set has been subdivided into two sets with 150 events selected in the first set and remaining 21 events in the second set. While the larger set has been used to develop OSR relation, the smaller set has been used to test and validate the methodology adopted for using OSR for homogenization of earthquake catalog.

The OSR relation has been derived by assuming the error variance ratio of the two magnitude determinations equal to unity (Ristau, 2009) as follows:

$$M_w = 1.3(\pm 0.004)m_{b,proxy} - 1.4(\pm 0.13), \quad R^2 = 0.88 \text{ and } \sigma = 0.2 \quad (17)$$

The reason to use $m_{b,proxy}$ instead of $m_{b,obs}$ in this relation is the criterion of minimization of orthogonal residuals in OSR (see Fig. 2a). OSR predicts m_y from given m_x and m_y values. M_w value thus depends on both the values of $m_{b,obs}$ as well as $M_{w,obs}$ which is shown in Table 1. For example, for the event # 124, M_w value

Table 2

Comparison of estimation of M_w values obtained using $m_{b,proxy}$ (obtained from OSR relation and Eq. (18)) values with the values obtained through projection of ($m_{b,obs}$, $M_{w,obs}$) on the OSR line for 21 test data points for Northeast India and adjoining region. The header abbreviations stand for MO – Month, DD – Day, HH – hour, MM – minute, SS – second, Lat – latitude, Long – longitude.

Event no.	Year	MO	DD	HH	MM	SS	Lat (°N)	Long (°E)	$m_{b,obs}$	$M_{w,obs}$	Projection of observed point on OSR line M_w	M_w (using Col 9 Eqs. (17) and (18))	Diff (12)–(11)
											(11)	(12)	(13)
1	2004	10	8	21	48	3.7	24.1738	94.07	4.8	4.8	4.8	4.9	0.1
2	2004	12	9	8	48	59	24.715	92.54	5.4	5.3	5.4	5.6	0.2
3	2005	1	18	3	2	53	22.9416	94.6	4.9	4.8	4.9	5.0	0.1
4	2005	2	3	20	13	29	26.1542	95.51	5.0	4.9	5.0	5.2	0.2
5	2005	2	15	11	15	5.8	24.5336	92.38	5.1	5.0	5.1	5.3	0.2
6	2005	2	15	13	5	48	24.4056	94.45	5.1	5.2	5.2	5.3	0.1
7	2005	3	23	5	59	6.4	26.0632	95.18	4.9	4.9	4.9	5.0	0.1
8	2005	3	25	13	34	38	25.4596	94.77	5.2	5.2	5.3	5.4	0.1
9	2005	3	26	20	32	9.7	28.1938	87.86	4.8	4.7	4.8	4.9	0.1
10	2005	6	1	20	6	40	28.8245	94.58	5.9	5.8	6.0	6.2	0.2
11	2005	7	17	1	4	43	21.0259	94.98	5.0	5.0	5.0	5.2	0.2
12	2005	8	20	12	50	49	31.2773	88.09	5.0	4.9	5.0	5.2	0.2
13	2005	9	18	7	25	58	24.6147	94.75	5.5	5.7	5.7	5.7	0.0
14	2005	12	29	7	20	54	24.9509	96.11	4.8	5.1	5.0	4.9	–0.1
15	2006	2	14	0	55	25	27.3867	88.42	5.3	5.3	5.4	5.5	0.1
16	2006	2	21	15	10	22	31.726	95.02	4.7	4.9	4.8	4.8	0.0
17	2006	2	23	20	4	55	26.9581	91.71	5.4	5.4	5.5	5.6	0.1
18	2006	4	19	21	5	41	31.5859	90.45	5.1	5.7	5.5	5.3	–0.2
19	2006	5	11	17	22	55	23.3591	94.27	5.7	5.6	5.8	6.0	0.2
20	2006	6	4	2	24	18	30.7195	98.97	4.7	4.9	4.8	4.8	0.0
21	2006	8	12	20	46	12	24.7088	92.74	4.8	5.0	4.9	4.9	0.0

on OSR line for the data pair (5.5, 5.0) is 5.3 but the M_w value becomes 5.6 for the data pair (5.5, 5.5) in event #147 as given in Table 1. Therefore, in order to obtain the M_w value predicted by the OSR relation for any observed value of $m_{b,obs}$ of the catalog, the corresponding $m_{b,proxy}$ value is required to be determined which can be directly substituted in the derived OSR relation (Eq. (17)). The procedure is also shown in Fig. 2a where the $m_{b,obs}$ 5.5 is projected to corresponding point on the OSR line with $m_{b,proxy}$ of 5.1 and $M_w = 5.3$.

In this regard, $m_{b,proxy}$ values are first obtained for the given 150 data pairs ($m_{b,obs}$, $M_{w,obs}$) by orthogonally projecting on the OSR line. Then, a standard linear relation is obtained based on 150 pairs of ($m_{b,obs}$, $m_{b,proxy}$) as follows:

$$m_{b,proxy} = 0.878(\pm 0.03)m_{b,obs} + 0.653(\pm 0.15),$$

$$R^2 = 0.72 \text{ and } \sigma = 0.17 \quad (18)$$

The above relation between $m_{b,obs}$ and $m_{b,proxy}$ is shown in Fig 2b.

On substituting $m_{b,proxy}$ value given by Eq. (18) in Eq. (17), the M_w value corresponding to a $m_{b,obs}$ value is obtained. Using this procedure, predicted M_w value for any $m_{b,obs}$ value contained in a catalog is obtained. To analyze the effect of estimation of M_w from $m_{b,proxy}$, correlation coefficients for the two series are estimated for 150 events. The increase in correlation coefficient of 0.84 between $m_{b,obs}$ and $M_{w,obs}$ to 0.94 between $m_{b,obs}$ and M_w estimated using $m_{b,proxy}$ implies the suppression of errors.

An endeavor has been made to validate the results of the proposed procedure of finding M_w values from Eqs. (17) and (18), using a test sample (not used in deriving the OSR relation) of 21 events data with $m_{b,obs}$ and $M_{w,obs}$ values. The M_w estimated using above methodology are found to be in good agreement with the M_w values of the orthogonal projections of the given data pairs (Table 2). The procedure developed in this study can be used for conversion of magnitudes having measurement errors to a unified moment magnitude to obtain a homogenized catalog.

5. Discussion and conclusion

Different magnitude determination procedures in use involve measurement errors which vary from one magnitude type to other. Most studies dealing with the homogenization of catalog perform conversion of magnitudes (e.g., M_L , m_b , M_s) to unified moment magnitude M_w through scaling relations which, in general, are of SR type. However, if both the magnitudes involved have measurement errors, which is the case with almost all the magnitude scales, the use of SR procedure may lead to incorrect results. In such a situation, it is better to use OSR which takes into account the errors on both the magnitudes (Stromeyer et al., 2004; Thingbaijam et al., 2008; Ristau, 2009).

In this study, we demonstrate the applicability of OSR for conversion of body wave magnitudes to M_w for the purpose of a homogenized earthquake catalog. An OSR relationship (Eq. (17)) has been developed based on a data set of 150 events with $m_{b,obs}$ and $M_{w,obs}$ values considered from ISC and GCMT databases for Northeast India and adjoining region for the period 1978–2006. For each of the observed data point ($m_{b,obs}$, $M_{w,obs}$), the corresponding point on the OSR line is to be determined as predicted M_w and the $m_{b,proxy}$ values. Therefore, a linear relation (Eq. (18)) has been obtained between the observed $m_{b,obs}$ and $m_{b,proxy}$ values based on 150 data pairs of values. This relation can be used to determine the $m_{b,proxy}$ value corresponding to any given $m_{b,obs}$ value. The corresponding M_w value is then obtained by directly substituting the $m_{b,proxy}$ value into the OSR relation. The improvement in the correlation coefficient value between $m_{b,obs}$ and M_w estimated using $m_{b,proxy}$ compared to the

value between $m_{b,obs}$ and $M_{w,obs}$ reveals the correct use of OSR relationship for homogenization. The OSR procedure developed in this study can be used to homogenize any catalog containing different magnitude types (e.g., M_L , m_b , M_s) having measurement errors, by their conversion to unified magnitude M_w . The proposed procedure also remains valid in case the magnitudes have measurement errors of different orders, i.e. the error variance ratio is different from unity.

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